

1. AIM

To design an CMOS inverter repeater using minimum channel length and minimum width in 180nm technology and determine

- 1] R_{out} , (output resistance of the inverter)
- 2] C_{out} , (output capacitance of the inverter)
- 3] intrinsic delay t_p , (intrinsic delay of the inverter)
- 4] R_n , R_p (resistance of NMOS and PMOS)
- 5] k (number of stage of repeaters)

to calculate propagation delay without inverter/buffer insertion and with inverter/buffer insertion and to compare the results.

2. TOOL

Cadence Virtuoso

3. THEORY

A long interconnect has significant resistance R_{int} and capacitance C_{int} , which increase the propagation delay of the CMOS inverter driving it.

A repeater is an intermediate CMOS inverter inserted in the interconnect to divide it into smaller sections. This reduces the wire delay by allowing each inverter to drive a smaller portion of the interconnect.

The interconnect is modeled as a distributed RC network, with its delay approximated as $t_{wire} = 0.4R_{int}C_{int}$, while the inverter and other components are represented using a lumped RC model with delay $t_{inv} = 0.7R_oC_o$.

Without repeater insertion, the complete interconnect is driven by a single inverter, resulting in a large RC propagation delay. With k repeater stages, the interconnect is divided into k sections and the total delay is obtained by summing the wire and inverter delays.

Increasing the number of repeaters reduces the wire delay but adds inverter delay. Therefore, an optimum number of stages k exists that minimizes the total propagation delay.

The delays with and without repeater insertion are calculated and compared to determine the improvement obtained by using CMOS repeaters.

4. TABLES

Table 4.1: Design Parameters

given parameter	value	unit
technology	180	nm
V_{dd}	1.8	V
V_{tn}	0.45	V
$ V_{tp} $	0.45	V
minimum width of nmos	400	nm
minimum width of pmos	800	nm
k_n'	70	$\mu\text{A}/\text{V}^2$
k_p'	20	$\mu\text{A}/\text{V}^2$
R_{int}	68	$\text{k}\Omega$
C_{int}	60	fF

Table 4.2: Derived and observed intrinsic parameters

derived and observed parameter	value	unit
t_{phl}	9.8587	s
t_{plh}	27.33919	ps
t_p	0.86	ns
R_n	4.762	$\text{k}\Omega$
R_p	16.67	$\text{k}\Omega$
R_{out}	10.716	$\text{k}\Omega$
C_n	3.0004	fF
C_p	2.3768	fF
C_{out}	3.0004	fF
k	8	unitless

Table 4.3: Result

parameter	without inv/buf insertion	with inv/buf insertion	unit	error
t_{phl}	2.882	1.17325	ps	-59.2904%
t_{plh}	3.119	1.23534	ps	-60.3931%
t_p	3.0005	1.2043	ps	-59.8634%

5. CALCULATION

$$R_n = 1000 / (k_n * (W/L) * (V_{dd} - V_{tn}))$$

$$= 1000 / (70 * (400/180) * (1.8 - 0.45)) = 4.7619048k\Omega \approx 4.762k\Omega$$

$$R_p = 1000 / (k_p * (W/L) * (V_{dd} - V_{tp}))$$

$$= 1000 / (20 * (400/180) * (1.8 - 0.45)) = 16.6666667k\Omega \approx 16.67k\Omega$$

$$R_{out} = (R_n + R_p) / 2$$

$$= (4.7619048 + 16.6666667) / 2 = 10.7142858k\Omega \approx 10.714k\Omega$$

$$C_{out1} = t_{phl} / (0.69 * R_n)$$

$$= 9.8587 / (0.69 * 4.7619048) = 3.0004739fF \approx 3fF$$

$$C_{out2} = t_{plh} / (0.69 * R_p)$$

$$= 27.33919 / (0.69 * 16.6666667) = 2.3773209fF \approx 2.3773fF$$

$$C_{out} = \max(C_{out1}, C_{out2})$$

$$= \max(3.0004739, 2.3773209) = 3.0004739fF$$

$$t_p = (t_{phl} + t_{plh}) / 2$$

$$= (9.8587 + 27.33919) / 2 = 18.598945ps \approx 18.5989ps$$

$$k = \sqrt{(R_{int} * C_{int} * 0.4) / (R_{out} * C_{out} * 0.7)}$$

$$= \sqrt{(68 * 60 * 0.4) / (10.7142858 * 3.0004739 * 0.7)} = 8.5159777 \approx 8$$

$$t_{HL(no_rep)} = 3.119ns$$

$$t_{LH(no_rep)} = 2.882ns$$

$$t_{p(no_rep)} = (t_{phl_no_rep} + t_{plh_no_rep}) / 2$$

$$= (3.119 + 2.882) / 2 = 3.0005ns$$

$$t_{HL(rep)} = 1.23534ns$$

$$t_{LH(rep)} = 1.17325ns$$

$$t_{p(rep)} = (t_{phl_rep} + t_{plh_rep}) / 2$$

$$= (1.23534 + 1.17325) / 2 = 1.204295ns \approx 1.2043ns$$

CIRCUIT DIAGRAMS & WAVEFORMS

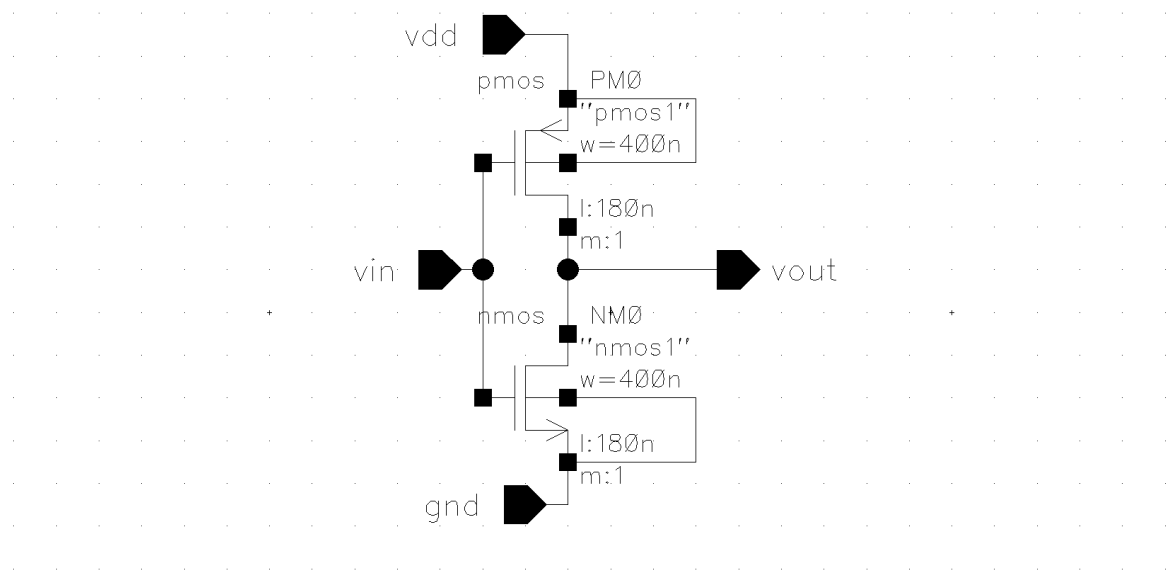


Figure .1: Schematic of minimum size inverter

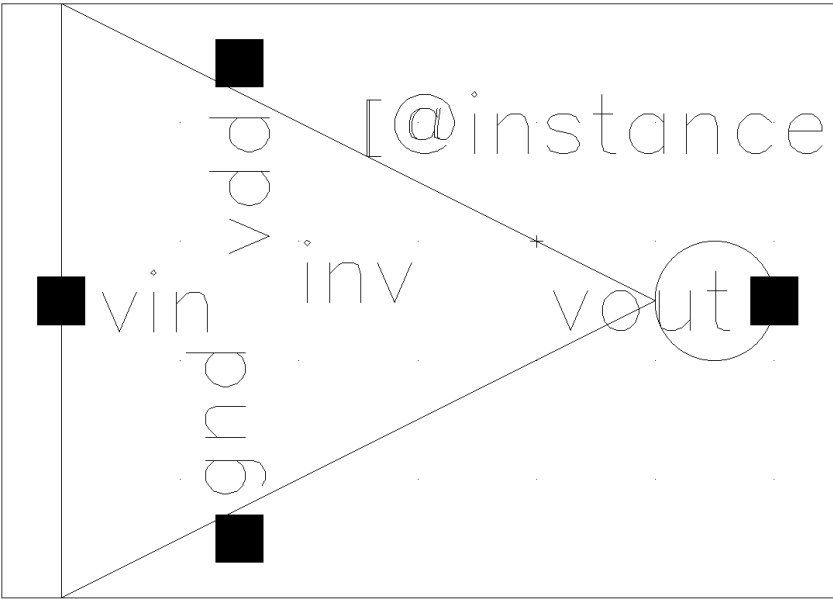


Figure .2: Symbol of minimum size inverter

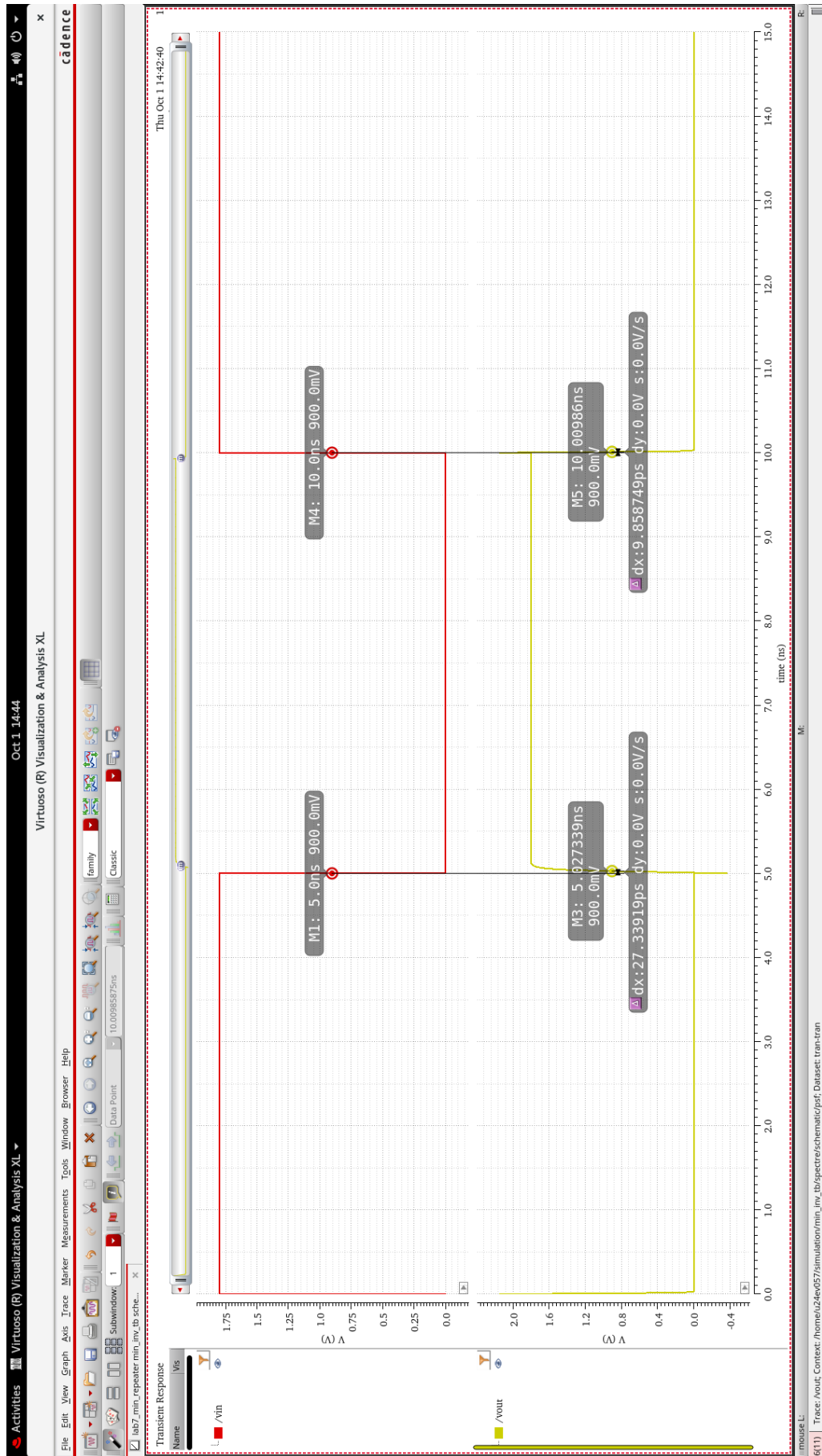


Figure .3: Transient waveform of minimum size inverter without load

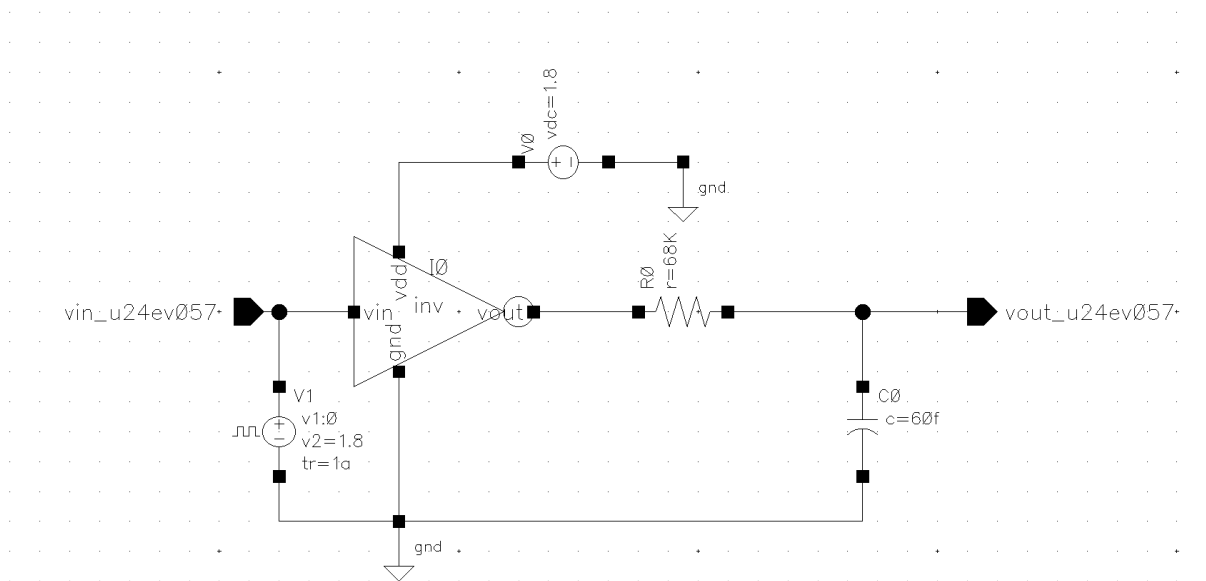


Figure .4: Schematic of inverter without inv/buf insertion (without repeaters)

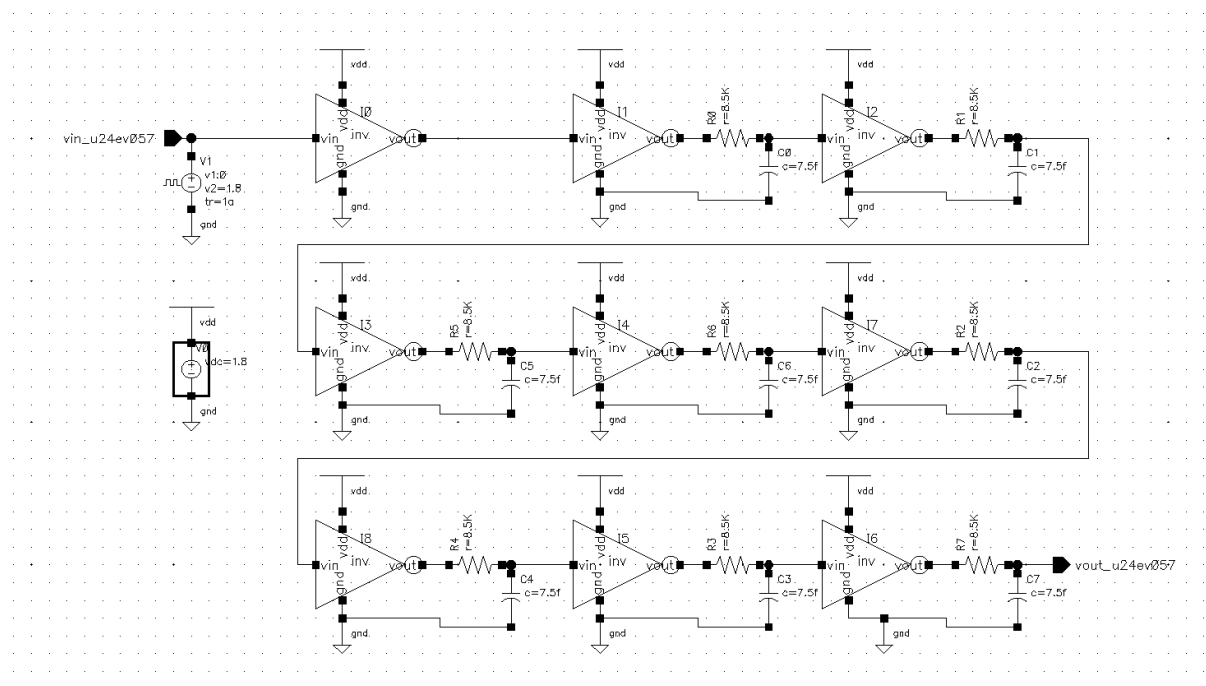


Figure .5: Schematic of inverter with inv/buf insertion (with repeaters)

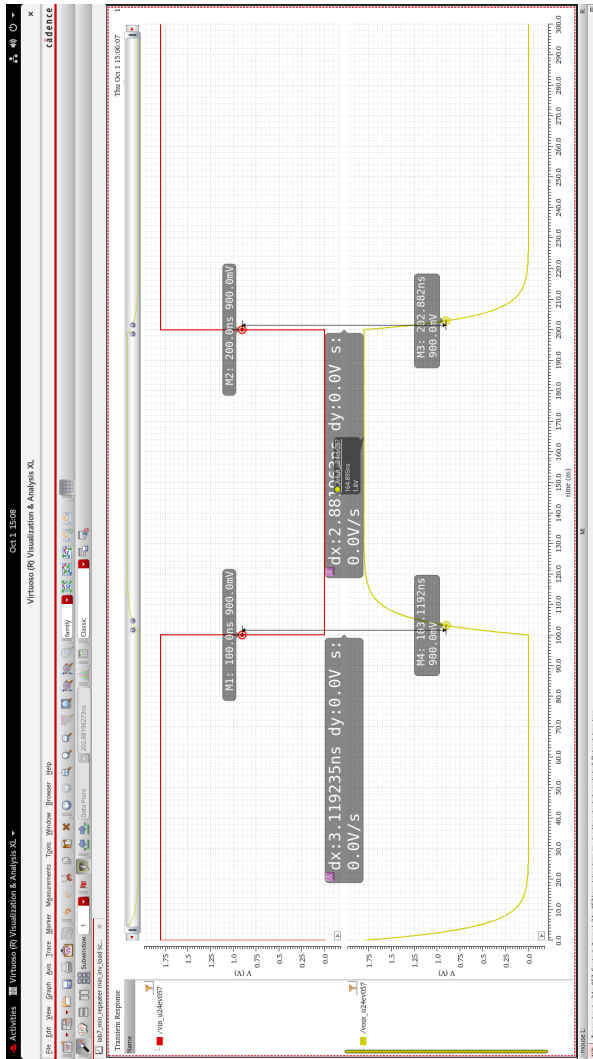


Figure .6: Transient waveform of inverter without inv/buf insertion (without repeaters)
 $t_{PHL} = 3.119\text{ns}$, $t_{PLH} = 2.882\text{ns}$, $t_p = 3.000\text{ns}$

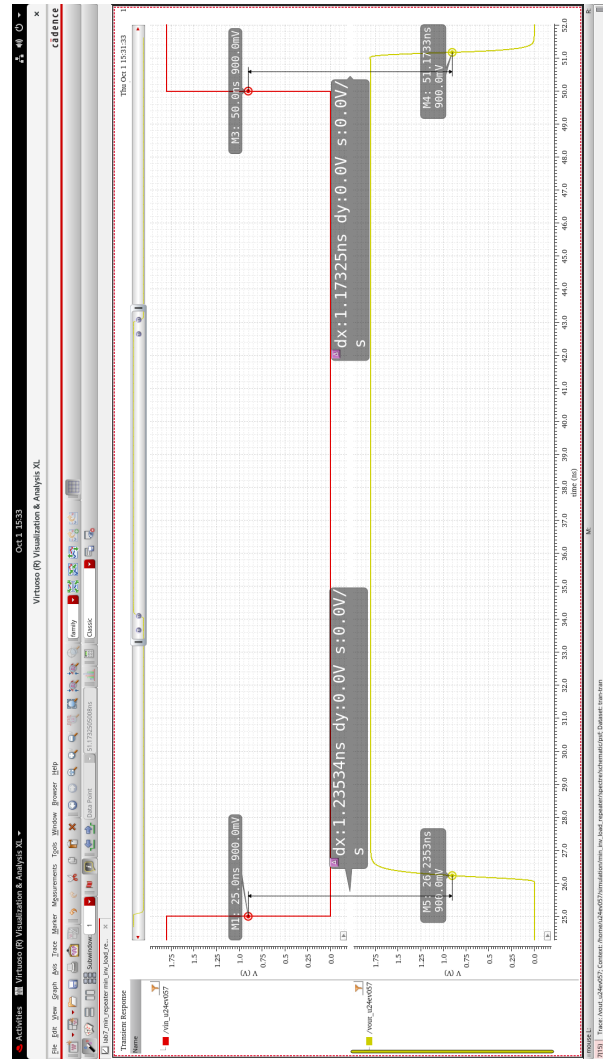


Figure .7: Transient waveform of inverter with inv/buf insertion (with repeaters)
 $t_{PHL} = 1.2353\text{ns}$, $t_{PLH} = 1.1732\text{ns}$, $t_p = 1.2043\text{ns}$

7. RESULT

The CMOS inverter repeater was successfully designed and simulated in 180 nm technology. Repeater insertion reduced the average propagation delay from approximately 3.00 ns to 1.20 ns, showing that repeaters significantly improve the speed of circuits driving large interconnects